MIL-T-12076A
19 JANUARY 1953
SUPERSEDING
MIL-T-12076(CMLC)
6 JUNE 1952

MILITARY SPECIFICATION

TIN, STRIP AND SHEET

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers two grades of tin in strip and sheet form.

1.2 Classification.—The material shall be furnished in the following grades, as specified (see 6.2):

Grade A—Pure.
Grade B—Commercial.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151 —Metals; General Specification for Inspection of.
UU-P-268 —Paper; Kraft, Wrapping.
UU-T-116 —Tape; Paper, Gummed, Water-Resistant.

MILITARY

JAN-P-106 —Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
MIL-L-10547 —Liners, Case, Waterproof.

U. S. ARMY

19-10 —Fillers, Interior Packing, Fiberboard, Corrugated, Single and Double Faced (For General Packing Purposes).

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Dimensions.—The thickness and width shall be as specified (see 6.2). A tolerance of plus or minus 10 percent shall be allowed on the specified thickness.

3.2 Workmanship.—The material shall be free from breaks, holes, and other imperfections adversely affecting its suitability for the purpose intended.

3.3 Purity.—The minimum tin content of grades A and B shall be 99.8 and 98.0 percent respectively, when tested as specified in 4.4.2.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of all the material from one grade offered for delivery at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:18Z
Check the source to verify that this is the current version before use.

4.2 Sampling.—Take a representative specimen weighing approximately 10 grams from each of 5 rolls selected at random from the lot. Place the specimens in a clean, dry container, seal and label to identify the sample with the lot represented. Test each specimen as specified in 4.4.2.

4.3 Inspection.

4.3.1 *Packaging, packing, and marking.*—The inspector shall inspect the packaging, packing, and marking for compliance with section 5.

4.3.2 *Procedures.*—For Air Force purchases, the general inspection procedures shall be in accordance with Specification QQ-M-151.

4.3.3 *Material.*—The material shall be inspected for defects and not less than three pieces selected by the inspector from each lot shall be measured. At least 10 measurements of thickness shall be made with a micrometer, the average of which shall be reported.

4.4 Tests.

4.4 *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determination shall be run in parallel with each test, using the same quantities or reagents used in the test, and corrections shall be applied when significant.

4.4.2 *Tin.*—Weigh, to the nearest 0.1 milligram (mg.) approximately 0.5 gram (gm.) of the specimen and dissolve the weighed portion in 25 milliliter (ml.) of concentrated hydrochloric acid by heating in a 500-ml. covered beaker. Dilute the solution to 100 ml. with cold water; add methyl red indicator and neutralize with ammonium hydroxide. Add approximately 50 ml. of saturated solution of ammonium nitrate and dilute to 300 ml. with boiling water, stirring the contents constantly. Allow the beaker to stand on a steam bath until the precipitate has settled and then decant the supernatant liquor through a hard filter paper. Wash the precipitate six times by decantation, with 100 ml. of boiling 1-percent nitric acid. Allow the precipitate to settle thoroughly after each washing. After the precipitate has been completely transferred to the filter paper, thoroughly wash and allow to drain, transfer to a porcelain crucible, dry on a hot plate, ignite at a low temperature until the filter paper has been consumed, and finally to constant weight. Calculate the percent tin as follows:

$$\text{Percent tin} = \frac{78.77 \times A}{B}$$

Where: A = weight of stannic oxide in gm.
B = weight of specimen in gm.

4.5 Rejection and resubmission.—If any specimen fails to conform to this specification, the lot shall be rejected. The contractor, at no expense to the Government, shall have the option of having an analysis and resubmitting the remaining portion of the lot for acceptance testing. If any specimen taken from the resubmitted portion of the lot fails to conform to this specification, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—The tin shall be rolled on tubes. Each roll shall be wrapped with kraft wrapping paper conforming to Specification UU-P-268, and sealed with water-resistant gummed tape conforming to Specification UU-T-116.

5.2 Packing.—The tin, packaged as specified in 5.1, shall be firmly and evenly packed in such a manner as to prevent damage to the ends of the rolls, in a nailed wood box conforming to style 2, 2½, or 4 of Specification JAN-P-106. A uniform number of rolls shall be packed in each box, and the gross weight shall be limited to 70 pounds. Pads of double-faced corrugated fiberboard conforming to Specification 19-10 shall be placed on

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:18Z
Check the source to verify that this is the current version before use.

bottom and top of the contents (to provide a tight pack within the box) and between tiers. For oversea shipments, boxes shall be provided with waterproof liners conforming to Specification MIL–L–10547, and sealed in accordance with the appendix thereof.

5.3 Marking.

5.3.1 *Packages.*—Each individual roll shall be legibly marked or labeled to properly identify the contents.

5.3.2 *Containers.*—In addition to any special marking required by contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129. The nomenclature shall read as follows:

Tin.
Grade _____*_____
Dimensions _____*_____
Specification MIL–T–12076

5.3.3 *Application.*—Unless otherwise specified, the packaging, packing and marking requirements specified herein apply only to direct purchases by or direct shipment to the Government.

6. NOTES

6.1 Intended use.—The material covered by this specification is intended for packaging purposes and for use in the assembly of ammunition and ammunition components.

* Applicable data to be entered by the contractor (see 6.2).

6.2 Ordering data.—Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Grade (see 1.2 and 3.3).
(c) Thickness and width (see 3.1).
(d) Linear feet or pounds required.

6.3 Sampling and testing.—When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its direction, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Air Force

Other Interest:
Army—T.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T22:18Z
Check the source to verify that this is the current version before use.